

E-Commerce Technology Store

Condensed Business Requirements Document

Condensed from the original Business / Functional Requirements documentation. Scope and business meaning are preserved; repetitive explanations and duplicated rules are consolidated.

1. Product Requirements

1.1 Product Information

Each product contains: Product ID, Name, Description, Price, Stock Quantity, Image, Brand, Category/Categories, IsActive, CreatedAt, and UpdatedAt. A separate Type field is not currently required. Products are associated with a Brand and one or more relevant Categories. IsActive allows Admins to make a product unavailable without deleting its information; timestamps track creation and modification. Database representation is deferred to Technical Design.

1.2 Permissions

Customers may view products/details, search, filter, sort, and add products to the cart, but cannot manage product information. Admins may create/update products, manage stock, and deactivate products.

1.3 Validation & Business Rules

Product Name is required; Price > 0; Stock cannot be negative; Brand and at least one Category must be valid; required data cannot be empty. Only Admins may create, update, or deactivate products. Inactive products cannot be purchased. Products with order history should be deactivated rather than permanently removed in a way that breaks historical orders. Customer-facing information should represent current active products.

1.4 Edge Cases

Inactive products cannot be purchased/added to cart. Invalid Brand/Category or invalid product data (e.g., negative price/stock or missing required fields) must be rejected. Deleting a product with order history must not break historical information; deactivation is preferred.

1.5 Product Specifications

Specifications vary by category and only relevant specifications should be shown. Examples: Laptop-CPU, RAM, Storage, GPU, Screen Size; Smartphone-RAM, Storage, Camera, Battery, Screen Size; Tablet-RAM, Storage, Battery, Camera, Screen Size; Monitor-Screen Size, Resolution, Refresh Rate, Panel Type; Keyboard-Size/Layout, Switch Type, Connection, Backlight; Headphones-Connection Type, Battery Life, Driver Size, Noise Cancellation; Microphone-Type, Connection, Polar Pattern, Frequency Response; Accessories-relevant product-specific specifications. Size is not mandatory for every product. Exact specification storage is deferred to Technical Design.

2. Product Details

2.1 Page Content

The Product Details Page must show Product Images, Name, Brand, Category/Categories, Description, Price, applicable Discount and Final Price, Stock Availability, relevant Specifications, Quantity Selector, and Add to Cart.

2.2 Stock Display

If Stock > 3, no low-stock message is needed. For 3, 2, or 1 units, show "Only 3/2/1 left in stock." For 0, show "Out of Stock" and disable/unavailable Add to Cart. The indicator is for customer awareness without exposing high inventory quantities.

2.3 Quantity & Add to Cart

Customers may select a quantity, but cannot select/add more than current stock. Add to Cart validates availability, rejects inactive/out-of-stock products, stores the selected quantity, and prevents unavailable quantities.

2.4 Discounts

Products may have discounts. When active, show original price, discount, final price, and optionally a discount badge. Without an active discount, show only the normal price. Final pricing must be reliable and not depend only on frontend calculations. Exact discount implementation is deferred.

2.5 Availability & Edge Cases

Availability states are Available, Low Stock (≤ 3), Out of Stock, and Inactive. If stock changes while the page is open, availability is revalidated before adding. If requested quantity exceeds stock, reject it and show the available quantity. Inactive products cannot

be purchased. If a discount expires, remove discount presentation and show the current price.

2.6 UX

The page should be clear, professional, easy to scan, focused on relevant specifications and pricing, transparent about discounts/availability, and make Add to Cart easy to find.

3. Product Discovery

3.1 Search

Customers can search by Product Name, Brand, Category, relevant Specifications, and Admin-defined keywords. Search should support relevant variations/alternative terms and return relevant products. Examples include Laptop, PC, Gaming Laptop, Samsung, Lenovo, RTX 4060, and 16GB.

3.2 Admin Keywords

Admins may add relevant alternative keywords to improve discoverability (e.g., ASUS ROG Strix G16 → ASUS, ROG, G16, Gaming Laptop, RTX 4060). Keywords must remain relevant and not misleading.

3.3 Categories & Brands

Initial categories include Laptops, Smartphones, Tablets, Monitors, Keyboards, Mice, Headphones, Microphones, PC Components, and Accessories. PC Components may include CPUs, GPUs, RAM, SSD, HDD, Motherboards, Power Supplies, PC Cases, and Cooling. Exact catalog is deferred. Brand filters should depend on category, focus initially on globally recognized brands, and include "Other" as a filter option without replacing actual brand data.

3.4 Filtering & Sorting

Filters should be dynamic by category and support multiple attributes simultaneously, e.g. Lenovo + 16GB + RTX 4060 + price range. Sorting includes Price Low High, Price High Low, Newest, Most Popular, and Highest Discount. Search may prioritize relevance, e.g. "Gaming Laptop" should favor products matching both concepts.

3.5 Pagination & Flow

Results use pagination; page size is implementation-dependent. Search, category, filters, and sorting must persist across pages. Flow: Search/Category Products → Filters → Sort →

Pagination → Product Details. Discovery should be clear, fast, relevant, and useful even when the customer does not know the exact product name.

| 4. Authentication & Authorization

4.1 Supported Features

Customer registration/login/logout, Admin account creation and verification, role-based access, secure passwords, and JWT-based authentication.

4.2 Customer Registration

Registration requires Username, Email, Password, and Confirm Password. Username must be unique. Customer password: minimum 8 characters, letters and numbers, and no obvious weak passwords such as 12345678, 123456789, or password. Separate uppercase/lowercase and special-character requirements are not mandatory.

4.3 Customer Activation

Customer accounts become usable immediately after successful registration; no Super Admin approval is required.

4.4 Admin Accounts

Admin accounts require stronger passwords (minimum 8 characters, uppercase, lowercase, number, special character) and remain Pending until Super Admin review. Super Admin may Approve or Reject; only approved Admins may access the Admin Dashboard.

4.5 Super Admin

There is one dedicated Super Admin account. It is not publicly registerable, selectable by customers, or treated as a normal role during registration. It has the highest privileges and is responsible for Admin verification and sensitive system operations.

4.6 Login, Logout & Roles

Users do not select their role during login; the system derives it from the authenticated account. Roles are Customer, Admin, and Super Admin. Logout terminates the authenticated session/token according to the chosen mechanism.

4.7 JWT & Authorization

JWT is used for authentication and protected features. Authentication identifies the user; authorization determines allowed actions. Server-side role authorization must prevent

access through URL changes, modified frontend code, custom API requests, or manipulated request data. Exact JWT/token/security implementation is deferred to Technical Design.

| 5. Shopping Cart

5.1 Overview & Guest Cart

The cart supports Product, Quantity, Current Unit Price, Discount, and Item Subtotal. Customers and guests may add, remove, update quantities, view the cart, and see totals. Guests must log in before final checkout/order creation.

5.2 Guest-to-Customer Merge

When a guest logs in, the guest cart merges with the customer's existing cart. Same-product quantities are combined, subject to current stock; invalid merged quantities must be corrected/rejected through stock validation.

5.3 Cart Operations

Adding an available product validates availability and stock. The same product must not create duplicate cart items; adding it again increases the existing quantity. Removing deletes the item and recalculates totals. Quantity must be >0 and $\leq$ current stock.

5.4 Stock, Price & Discounts

If an item becomes out of stock, it remains visible with a clear warning and blocks checkout until resolved. Cart totals use current product prices. The cart shows original/current price, discount, item subtotal, cart subtotal, total discount, and final total. Important calculations are server-validated.

5.5 Checkout Validation

Before checkout, validate that products still exist, are active/available, quantities are valid, stock is sufficient, current prices and discounts are valid. Any failure must be explained and block continuation.

5.6 Persistence & Admin Queue

Customer carts remain associated with their accounts. Guest carts persist until login, clearing, or expiry according to final implementation rules. Successfully created orders enter the Admin order-management FIFO queue by creation time; the cart itself is not a FIFO queue. Queue implementation is deferred.

5.7 UX

The cart should clearly show product, image, quantity, price, discount, subtotals, totals, and availability, and explain why checkout is blocked when validation fails.

6. Checkout

6.1 Overview

Checkout is for authenticated customers to review the cart, provide customer/shipping information, select payment, and proceed to order creation. Guests must log in. The customer must see a clear order summary before confirmation.

6.2 Information

Order summary includes products/images, quantities, unit prices, discounts, item subtotals, total discount, and final total. Customer information includes Full Name, Email, and Phone Number. Shipping includes Country, City, Address, and Postal Code. Required fields are validated.

6.3 Payment Methods

Cash on Delivery requires no payment proof. Bank Transfer is a manual external transfer: the system displays store payment information (Account Name, Bank Name, Account Number/IBAN and other required details), the customer transfers externally, then uploads Payment Proof. No real online payment gateway is implemented. PayPal/other external gateway support is outside the current scope.

6.4 Payment Proof

Payment Proof is required only for Bank Transfer, is associated with the order, and must be reviewable by Admin. Uploading proof does not equal verification; Admin verification is required.

6.5 Validation & Confirmation

Before confirmation validate customer/shipping information, payment method, cart availability, stock, current prices, discounts, and required Bank Transfer proof. The customer must explicitly confirm the Order Summary, customer/shipping information, payment method, payment information/proof, and final total.

| 7. Order Creation

7.1 Creation & Order Number

An order is created after an authenticated customer completes Checkout and selects Place Order. Each order has a unique Order Number.

7.2 Final Validation

Immediately before creation, revalidate the cart, product existence/active status, quantities, stock, customer/shipping information, payment method, required proof, prices, and discounts. Server-side recalculation is mandatory; previous Cart/Checkout validation alone is insufficient.

7.3 Stock & Partial Availability

Stock is checked again at order placement. If requested stock is unavailable, the order is not created and the customer is informed. If one or more products become unavailable, do not silently remove them and place the order. Ask whether to continue with the remaining items; if yes, remove unavailable items, revalidate, recalculate, and retry; if no, create no order.

7.4 Pricing & Snapshot

Server recalculates unit prices, quantities, discounts, subtotals, and final total. Price is locked at order creation. Order items preserve Product ID, Product Name, Unit Price, Quantity, Discount, and Item Subtotal so history remains accurate if the product is later renamed, repriced, updated, deactivated, or removed.

7.5 Stock, Payments & Cart

After successful creation, stock is deducted/reserved. COD creates a Pending order with no payment verification. Bank Transfer creates Pending + Payment Status Pending Verification; Admin reviews proof, which may become Verified or Rejected. Rejection follows cancellation/rejection rules and returns reserved stock when applicable. Successfully ordered items are removed from the cart; failed creation leaves the cart available.

7.6 Atomicity & Failure Handling

Order creation must behave atomically across Order, Order Items, stock update, and cart clearing. Critical failures roll back related changes. No incomplete order should remain; applicable stock changes roll back, the cart remains available, a clear error is shown, and retry is possible.

| 8. Order Confirmation

8.1 Confirmation Page

After successful order creation, redirect to an Order Confirmation page with a clear success message and Order Number, Total Amount, Payment Method, and Order Status. Bank Transfer orders also show that payment verification is pending.

8.2 Actions & Reviews

Customer can View Order and Continue Shopping. Rating/review is optional and becomes available only after the order is Delivered; customers are not required to review.

8.3 UX

The page should immediately communicate that the order was created, which order it is, total cost, payment method, current order/payment status, where to view the order, and how to continue shopping.

| 9. Admin Order Management & Administration

9.1 Administration Structure

Admin handles daily store operations; Super Admin handles sensitive/system-level operations.

9.2 Admin Order Management

Admin can view all orders with Order Number, Customer, Order Date, Total, Payment Method, Payment Status, and Order Status; open full order details; confirm eligible orders; cancel eligible orders; and update status only through valid workflow rules. Invalid transitions/cancellations are rejected.

9.3 Bank Transfer Review

Admin can view, accept, or reject Bank Transfer Payment Proof. Acceptance sets Payment Status to Verified; rejection sets it to Rejected and triggers the applicable cancellation/rejection logic, including stock return where applicable.

9.4 Product & Keyword Management

Admin can create/update/deactivate products, update stock, normal product information, permitted categories/brands, prices, discounts, and regular product keywords/search information. All changes follow validation rules. Sensitive/system-level keyword configuration remains under Super Admin control.

9.5 Admin Restrictions

Admin cannot perform Super Admin operations. Server-side authorization must enforce this regardless of URL, frontend manipulation, custom requests, or modified data.

9.6 Super Admin

Super Admin can create, approve/reject, activate/deactivate, and manage Admin accounts/access; control sensitive system configuration; and oversee important administrative activities. Sensitive actions should be traceable; exact audit-log implementation is deferred.

9.7 Role Separation

Admin: daily store operations, orders, payment-proof review, products, stock, prices/discounts, normal administration. Super Admin: Admin accounts/approval/access, sensitive configuration, and high-level oversight.

| 10. Order Lifecycle & Status

10.1 Main Lifecycle

Pending → Confirmed → Processing → Shipped → Delivered. Pending is the initial status after successful order creation. Admin controls normal status operations through defined rules, and customers can view but not modify status.

10.2 Status Rules

Pending follows successful Checkout, final validation, stock validation, order creation, and Order Item creation. Confirmed means the Admin has confirmed an eligible order; Bank Transfer payment verification must satisfy payment rules before fulfillment. Processing means preparation has started. Shipped means the order has left the store for delivery. Delivered completes the normal lifecycle and enables eligible product rating/review.

10.3 Cancellation

Cancellation is a separate termination path and is allowed only where business rules permit. Eligible cancellation changes status to Cancelled, informs the customer, and returns reserved stock when applicable.

10.4 Payment Status

Payment Status is independent from Order Status. For Bank Transfer: Pending Verification → Verified or Rejected. Rejection may cause Order Cancellation and stock return; it is not itself an Order Status transition.

10.5 Invalid Transitions

Invalid transitions such as Pending → Shipped, Pending → Delivered, Confirmed → Delivered, Delivered → Processing, or Delivered → Pending must be rejected unless an explicit business rule permits them. Backend validation is required for every transition.

10.6 Responsibility & Visibility

Admin reviews, confirms, cancels, updates through the allowed workflow, and reviews/accepts/rejects payment proofs. Super Admin manages Admin accounts/approval and sensitive system operations. Customers always see current status in order details/history but cannot change it.

10.7 Status History & Stock

Important status changes should be recorded for traceability. Stock is deducted/reserved after successful order creation and returned after eligible cancellation/payment rejection. Duplicate stock restoration must be prevented.

10.8 Fraudulent Payment Proof

If intentionally fake/manipulated proof is confirmed: reject payment → cancel order → return stock → record incident. The customer account may be suspended, with a clear account-status message. Exact suspension implementation is deferred.

11. Cross-Cutting Business Rules & Deferred Technical Design

11.1 Core Rules

The system must protect historical order accuracy, prevent purchase of inactive/unavailable products, validate stock and pricing server-side, keep Payment Status separate from Order Status, require Admin verification for Bank Transfer, prevent invalid status changes, prevent duplicate stock restoration, and make delivered orders eligible for rating/review. Sensitive administrative actions should be traceable.

11.2 Deferred Technical Decisions

Across the document, the following remain for Technical Design: ERD/database structure and relationships; specification storage; API endpoints/design; backend/frontend architecture; search algorithm/ranking, indexing and query optimization; pagination implementation; JWT/token storage/expiration/refresh and password hashing; authentication/authorization middleware; user/role/account-status implementation; payment-proof storage/file validation; payment-status implementation; order/order-item structures; transactions, atomicity, concurrency and stock reservation/restoration; FIFO queue implementation; audit logging; account suspension mechanism; and exact UI/implementation details.

Document principle: Requirements are stated once in their most appropriate section. Later sections rely on those rules instead of repeating them.